



KTU NOTES

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**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPER**

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9. A stone is dropped from the top of a tower 70 m high. At the same time another stone is thrown up from the foot of the tower with a velocity of 30 m/s. Find the time taken for the two stones to cross each other from the top and after how much time the two stones cross each other.
10. State and explain the conditions of equilibrium of coplanar concurrent forces. (module 1)
11. Write down the expression for centroid of a rectangle about a line parallel to its base. (module 3)
12. Briefly explain the analysis of forces acting on a wedge with a horizontal base. (module 2)
13. A simply supported beam AB of span 4m is carrying point loads of 1kN, 2kN & 3kN at distances of 1m, 2m & 3m respectively from support A. Calculate reactions at support B. (module 2)
14. Define coefficient of friction and angle of friction. Establish a relation between them. (module 2)
15. Distinguish damped and undamped free vibrations (module 5)
16. A flywheel weighing 500N and having radius of gyration 0.4 m is brought to rest from 300rpm to 180 rpm in 1 minute. Calculate the torque acting on it. (module 5)
17. Differentiate between free vibration and forced vibration of bodies.
18. The position of a particle moving along a straight line is defined by the equation $S = t^3 + 3t^2 + 9t + 12$. Determine the time taken by the particle when its velocity becomes zero.
19. Three forces 20N, 30N and 40N act along AB, BC and CA respectively at the vertices of an equilateral triangle ABC. Find the resultant (module 1)
20. State and explain the principle of transmissibility of force with sketch.
21. Define free body diagram. (module 1)
22. State and explain Lami's theorem (module 1)
23. Define couple and explain its characteristics. With the help of a sketch, show how a force can be resolved into a force and a couple.
24. State the principle of transmissibility of rigid bodies.
25. F1 and F2 are two collinear forces. When they act in opposite directions, the resultant is 34N, when they act at right angles to each other, the resultant is 50N. Find F1 and F2. (module 1)

- Draw the free body diagram of the wedge and the block (module 2)
32. Distinguish between simple harmonic motion and periodic motion (
 33. Show that the acceleration at any instant during simple harmonic motion is proportional to the displacement from the mean position. (module 5)
 34. A flywheel of mass 5 tonne starts from rest and gets up a speed of 1000 revolutions per minutes. Find the average torque exerted on it if the radius of gyration is 1 m. (module 4)
 35. A force of magnitude 44 N acts through a point A (4,-1,7) in the direction $4\mathbf{i} + 6\mathbf{j} - 2\mathbf{k}$. Find the moment of the force about the point B (1,-3,2). (module 4)
 36. State D'Alembert's principle. Draw the free body diagram of a 1 kg mass moving upwards with an acceleration 'a' and also write the equilibrium equation using this principle (module 4)
 - 37.
 38. Define Radius of gyration, product of inertia and principal moment of inertia. (module 3)
 39. Define simple harmonic motion? Derive an expression for the acceleration of a particle executing simple harmonic motion. (module 5)
 40. Define stiffness and equivalent stiffness of spring. Give the expressions for the equivalent stiffness of two springs in series and parallel (module 5)
 41. A particle moves with simple harmonic motion. When it is at 1 m from the centre of the path, its velocities are 3 m/s and 2 m/s respectively. Find the period of the simple harmonic motion and the acceleration (module 5)
 42. Define the terms amplitude, period of oscillation and frequency in simple harmonic motion (module 5)
 43. A traffic signal of mass 500 N is hung with the help of two strings as shown in figure (Q1) below. Find the forces induced in the strings using Lami's theorem (module 4)

44. A force of 300N acts on a body of weight 500N. Find the acceleration using D'Alembert's principle. (module 4)
45. The angular acceleration of a particle $\alpha = 5t \text{ rad/s}^2$. Determine expression for velocity ' ω ' at any instant ' t ' of the motion, if the particle starts from rest.
- A force $\mathbf{F} = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$ is applied at the point B(1,-1.2). Find the moment of inertia of the force about a point A.
47. A solid circular disc of mass 10 kg and radius 0.3m is rotating about a horizontal axis through its center and perpendicular to the plane of the disc with a constant angular acceleration of 10 rad/s^2 . Determine the torque required to produce this acceleration. (module 5)
48. A body performing Simple Harmonic Motion has an amplitude of 0.1m and a period of oscillation 2.05 seconds. Find the velocity and acceleration of the body at the end of the first cycle after passing from mean positions.
- 49.

A system of parallel forces is acting on a rigid bar as shown in the figure. Find the resultant force and its position relative to point A.

